Week 9: Neural Network

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Motivation (1/2)

Given a dataset D,

x_1	x_2
0	1
2	3
4	5

How many coefficients θ_j are there for a polynomial of degree p? $\binom{n}{r} = \frac{n!}{r!(n-r)!}$, where n is the sum of the number of attributes and polynomial degrees, and r is the number of polynomial degrees.

Motivation (2/2)

How about this dataset,



Figure: A cat W100 x H100; obtained from istockphoto 1361394182

How many coefficients θ_j are there for a polynomial of degree p?

Artificial Neural Network

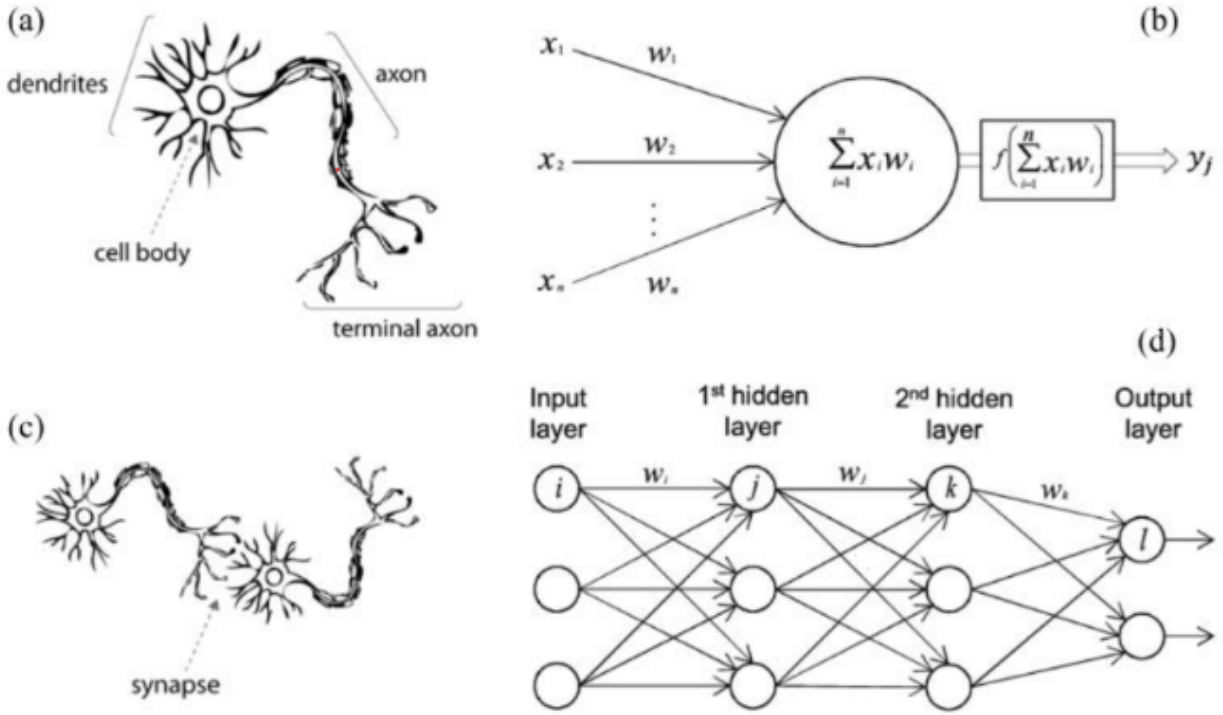


Figure 3. A biological neuron in comparison to an artificial neural network: **(a)** human neuron; **(b)** artificial neuron; **(c)** biological synapse; and **(d)** ANN synapses [39].

Figure: Artificial Neural Network

Artificial Neural Network

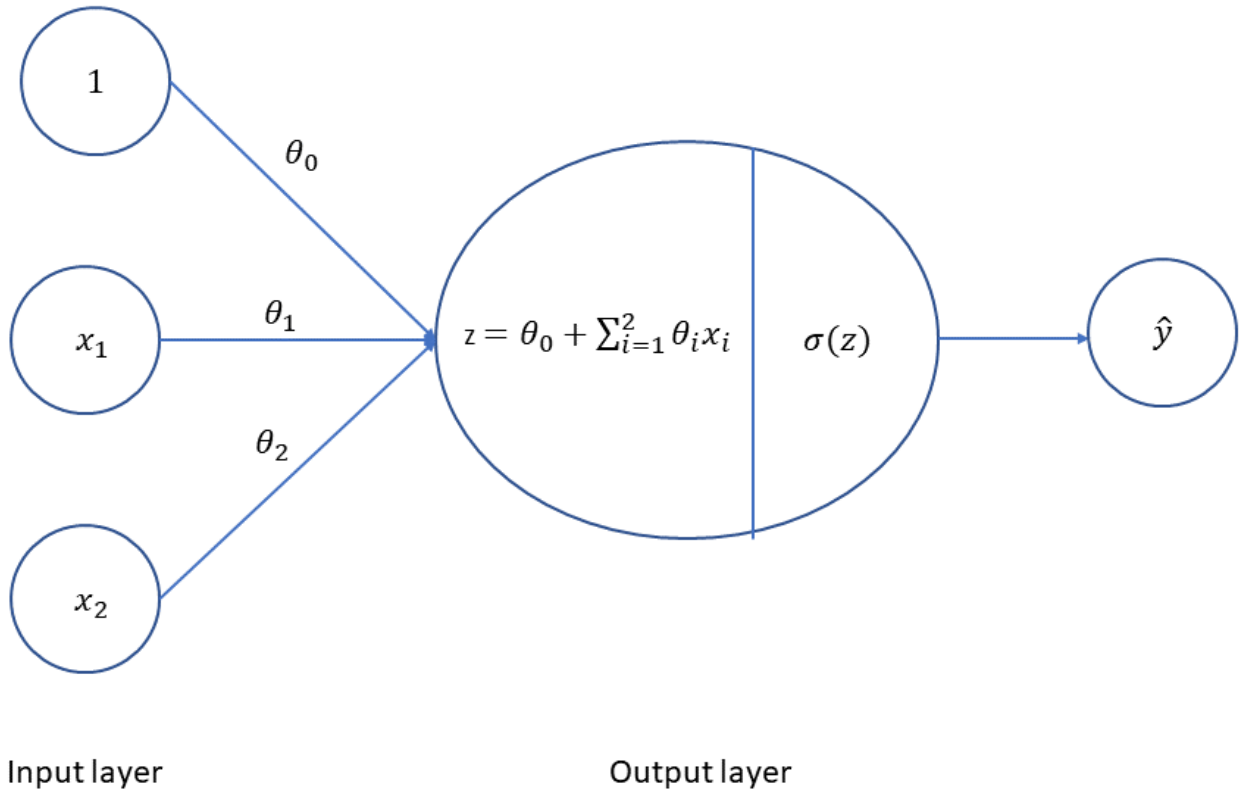
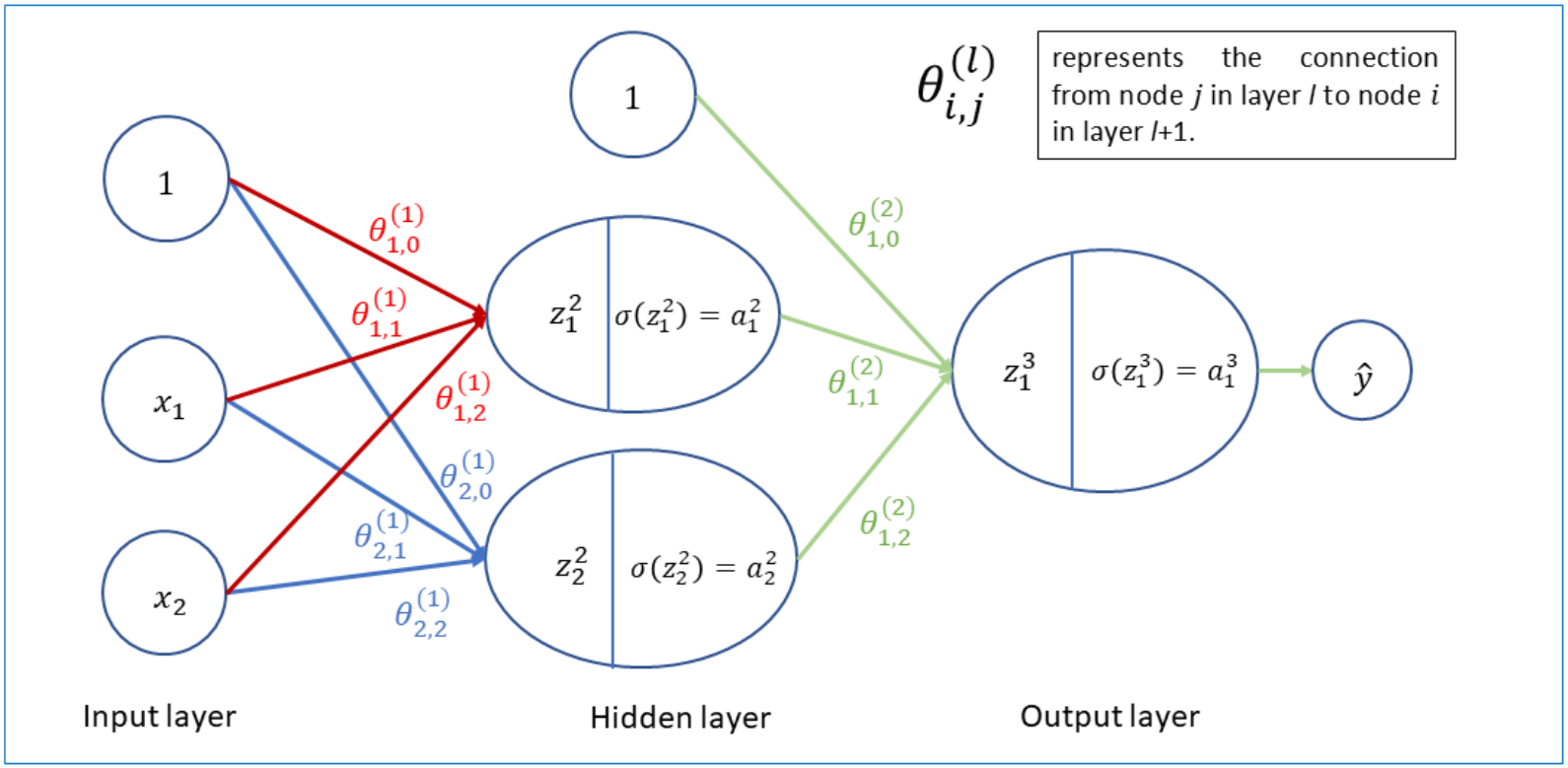


Figure: A simple Neural Network: Perceptron

Artificial Neural Network



$$a^2_1 = \sigma(\theta^{(1)}_{1,0} + \theta^{(1)}_{1,1}x_1 + \theta^{(1)}_{1,2}x_2)$$

$$a^2_2 = \sigma(\theta^{(1)}_{2,0} + \theta^{(1)}_{2,1}x_1 + \theta^{(1)}_{2,2}x_2)$$

$$a^3_1 = \sigma(\theta^{(2)}_{1,0} + \theta^{(2)}_{1,1}a^2_1 + \theta^{(2)}_{1,2}a^2_2)$$

$$\theta^{(1)} = \begin{bmatrix} \theta^{(1)}_{1,0} & \theta^{(1)}_{1,1} & \theta^{(1)}_{1,2} \\ \theta^{(1)}_{2,0} & \theta^{(1)}_{2,1} & \theta^{(1)}_{2,2} \end{bmatrix} \quad \theta^{(2)} = \begin{bmatrix} \theta^{(2)}_{1,0} & \theta^{(2)}_{1,1} & \theta^{(2)}_{1,2} \end{bmatrix}$$

Figure: Multilayer Perceptron

Training Data Example

AND Operation

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

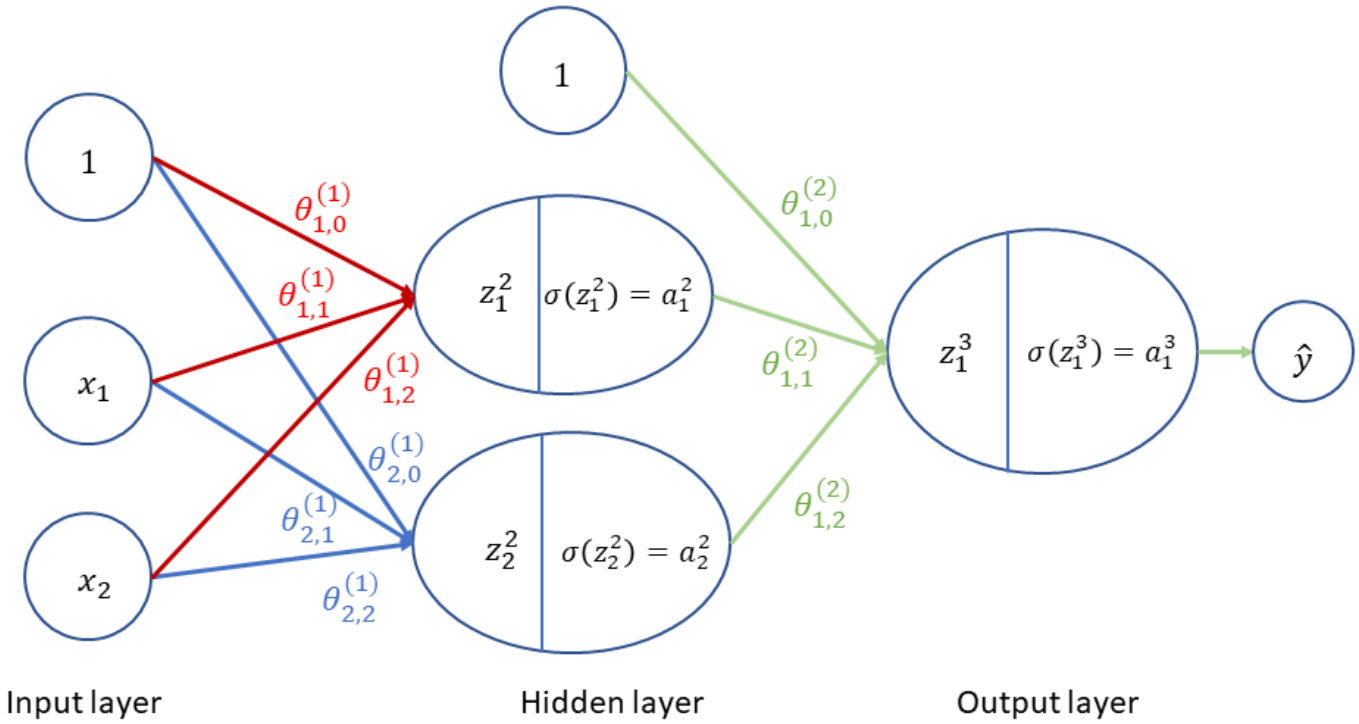
XNOR Operation

x_1	x_2	y
0	0	1
0	1	0
1	0	0
1	1	1

Training Process

- **Cost Function**: $J(\theta) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$
- **Goal**: $\theta^* = \arg\min J(\theta)$
- **For each sample in the training data:**
  - Step 1: Forward pass
    Calculate $\hat{y}$, i.e. a^L, where a big L is the output layer.
  - Step 2: Calculate loss of the output layer, i.e. $a^L - y$.
  - Step 3: Backward pass (Backpropagation)
    Find $\frac{\partial J(\theta)}{\partial \theta_{ij}}$
  - Step 4: Update each $\theta_{i,j}^{(l)}$ (weights)

Step 1&2: Forward pass & Error of last layer



Step 3: Backward pass (Backpropagation) 1/6

$\theta^{(2)}$

- $\frac{\partial J(\theta)}{\partial \theta_{1,0}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,0}^{(2)}}$
- $\frac{\partial J(\theta)}{\partial \theta_{1,1}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,1}^{(2)}}$
- $\frac{\partial J(\theta)}{\partial \theta_{1,2}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,2}^{(2)}}$

Step 3: Backward pass (Backpropagation) 2/6

$\theta^{(2)}$, **e.g.,** $\theta^{(2)}_{1,0}$

- $\frac{\partial J(\theta)}{\partial \theta^{(2)}_{1,0}} = \frac{\partial J(\theta)}{\partial a^3_1} \cdot \frac{\partial a^3_1}{\partial z^3_1} \cdot \frac{\partial z^3_1}{\partial \theta^{(2)}_{1,0}}$
- $\frac{\partial J(\theta)}{\partial a^3_1} = \frac{\partial(-y\log(a^3_1)-(1-y)\log(1-a^3_1))}{\partial a^3_1} = -\frac{y}{a^3_1} + \frac{1-y}{1-a^3_1} = \frac{1-y}{1-a^3_1} - \frac{y}{a^3_1}$
- $\frac{\partial a^3_1}{\partial z^3_1} = \frac{\partial \sigma(z^3_1)}{\partial z^3_1} = \sigma(z^3_1)\cdot(1-\sigma(z^3_1)) = a^3_1 \cdot (1-a^3_1)$
- $\frac{\partial z^3_1}{\partial \theta^{(2)}_{1,0}} = \frac{\partial(\theta^{(2)}_{1,0}+\theta^{(2)}_{1,1}.a^2_1+\theta^{(2)}_{1,2}.a^2_2)}{\partial \theta^{(2)}_{1,0}} = 1$
- So that, $\frac{\partial J(\theta)}{\partial \theta^{(2)}_{1,0}} = \left(\frac{1-y}{1-a^3_1} - \frac{y}{a^3_1}\right)\cdot\left(a^3_1\cdot(1-a^3_1)\right)\cdot 1$
  $= (a^3_1 - y)\cdot 1$

Step 3: Backward pass (Backpropagation) 3/6

- $\frac{\partial J(\theta)}{\partial \theta_{1,0}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,0}^{(2)}} = (a_1^3 - y) \cdot 1$
- $\frac{\partial J(\theta)}{\partial \theta_{1,1}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,1}^{(2)}} = (a_1^3 - y) \cdot a_1^2$
- $\frac{\partial J(\theta)}{\partial \theta_{1,2}^{(2)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial \theta_{1,2}^{(2)}} = (a_1^3 - y) \cdot a_2^2$
- Let say $\delta_1^3 = (a_1^3 - y)$ be the error of the first node in the last layer ($L = 3$ in this example).

Step 3: Backward pass (Backpropagation) 4/6

$\theta^{(1)}$

- $\frac{\partial J(\theta)}{\partial \theta_{1,0}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,0}^{(1)}}$
- $\frac{\partial J(\theta)}{\partial \theta_{1,1}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,1}^{(1)}}$
- $\frac{\partial J(\theta)}{\partial \theta_{1,2}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,2}^{(1)}}$
- $\frac{\partial J(\theta)}{\partial \theta_{2,0}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,0}^{(1)}}$
- $\frac{\partial J(\theta)}{\partial \theta_{2,1}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,1}^{(1)}}$
- $\frac{\partial J(\theta)}{\partial \theta_{2,2}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,2}^{(1)}}$

Step 3: Backward pass (Backpropagation) 5/6

$\theta^{(1)}$

- $\frac{\partial J(\theta)}{\partial \theta_{1,0}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,0}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,1}^{(2)} \cdot a_1^2(1 - a_1^2) \cdot 1$
- $\frac{\partial J(\theta)}{\partial \theta_{1,1}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,1}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,1}^{(2)} \cdot a_1^2(1 - a_1^2) \cdot a_1^1$
- $\frac{\partial J(\theta)}{\partial \theta_{1,2}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial \theta_{1,2}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,1}^{(2)} \cdot a_1^2(1 - a_1^2) \cdot a_2^1$
- $\frac{\partial J(\theta)}{\partial \theta_{2,0}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,0}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,2}^{(2)} \cdot a_2^2(1 - a_2^2) \cdot 1$
- $\frac{\partial J(\theta)}{\partial \theta_{2,1}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,1}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,2}^{(2)} \cdot a_2^2(1 - a_2^2) \cdot a_1^1$
- $\frac{\partial J(\theta)}{\partial \theta_{2,2}^{(1)}} = \frac{\partial J(\theta)}{\partial a_1^3} \cdot \frac{\partial a_1^3}{\partial z_1^3} \cdot \frac{\partial z_1^3}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial \theta_{2,2}^{(1)}} = (a_1^3 - y) \cdot \theta_{1,2}^{(2)} \cdot a_2^2(1 - a_2^2) \cdot a_2^1$

Note that a_1^1 *is* x_1, a_2^1 *is* x_2

Step 3: Backward pass (Backpropagation) 6/6

- Finally, we get $\frac{\partial J(\theta)}{\partial \theta_{i,j}^{(l)}} = \delta_i^{l+1} \cdot a_j^l$, with a condition that
  - if $l + 1 = L$ then $\delta_k^L = (a_k^l - y_k)$,

    where k is a class's index in y.
  - else if $l = L - 1, L - 2, ..., 2$ then $\delta_i^l = (\sum_{k=1}^{C} \delta_k^{l+1} \cdot \theta_{k,i}^{(l)}) \cdot a_i^l \cdot (1 - a_i^l)$,

    where C is a number of classes in y.

Step 4: Update each $\theta_{i,j}^{(l)}$ (weights)

- $\theta_{i,j}^{(l)} = \theta_{i,j}^{(l)} - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_{i,j}^{(l)}}$
- $\theta_{i,j}^{(l)} = \theta_{i,j}^{(l)} - \alpha \cdot (\delta_i^{l+1} \cdot a_j^l)$, where
  - $\theta_{i,j}^{(l)}$: The parameter (e.g., weight) connecting node (j) in layer (l) to node (i) in layer $l+1$.
  - α: The learning rate.
  - δ_i^{l+1}: The error term (delta) for node (i) in layer $l+1$.
  - a_j^l: The activation of node (j) in layer (l).

Activation Functions

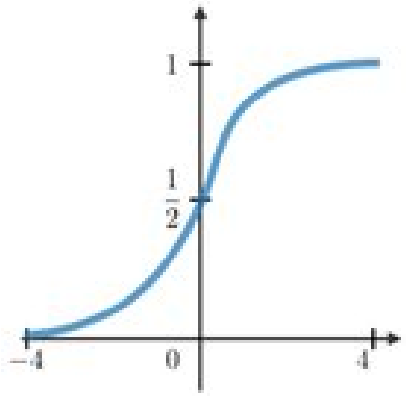
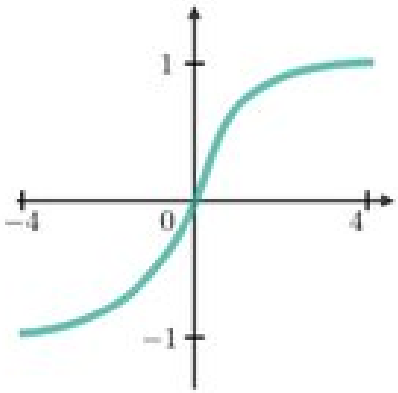
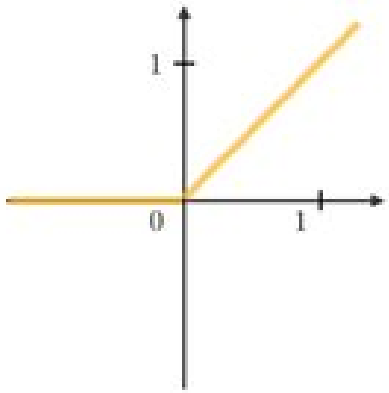
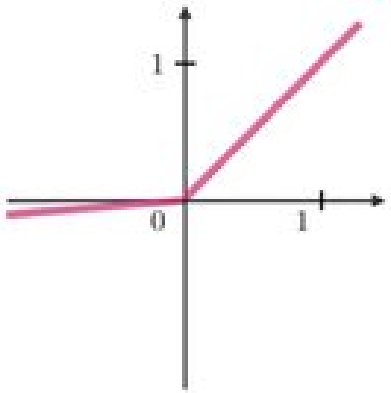
Sigmoid	Tanh	ReLU	Leaky ReLU
$g(z) = \frac{1}{1+e^{-z}}$	$g(z) = \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$	$g(z) = \max(0, z)$	$g(z) = \max(\epsilon z, z)$ with $\epsilon \ll 1$
			

Figure: Activation function Ref:
http://dx.doi.org/10.13140/RG.2.2.14692.60800

Softmax Functions

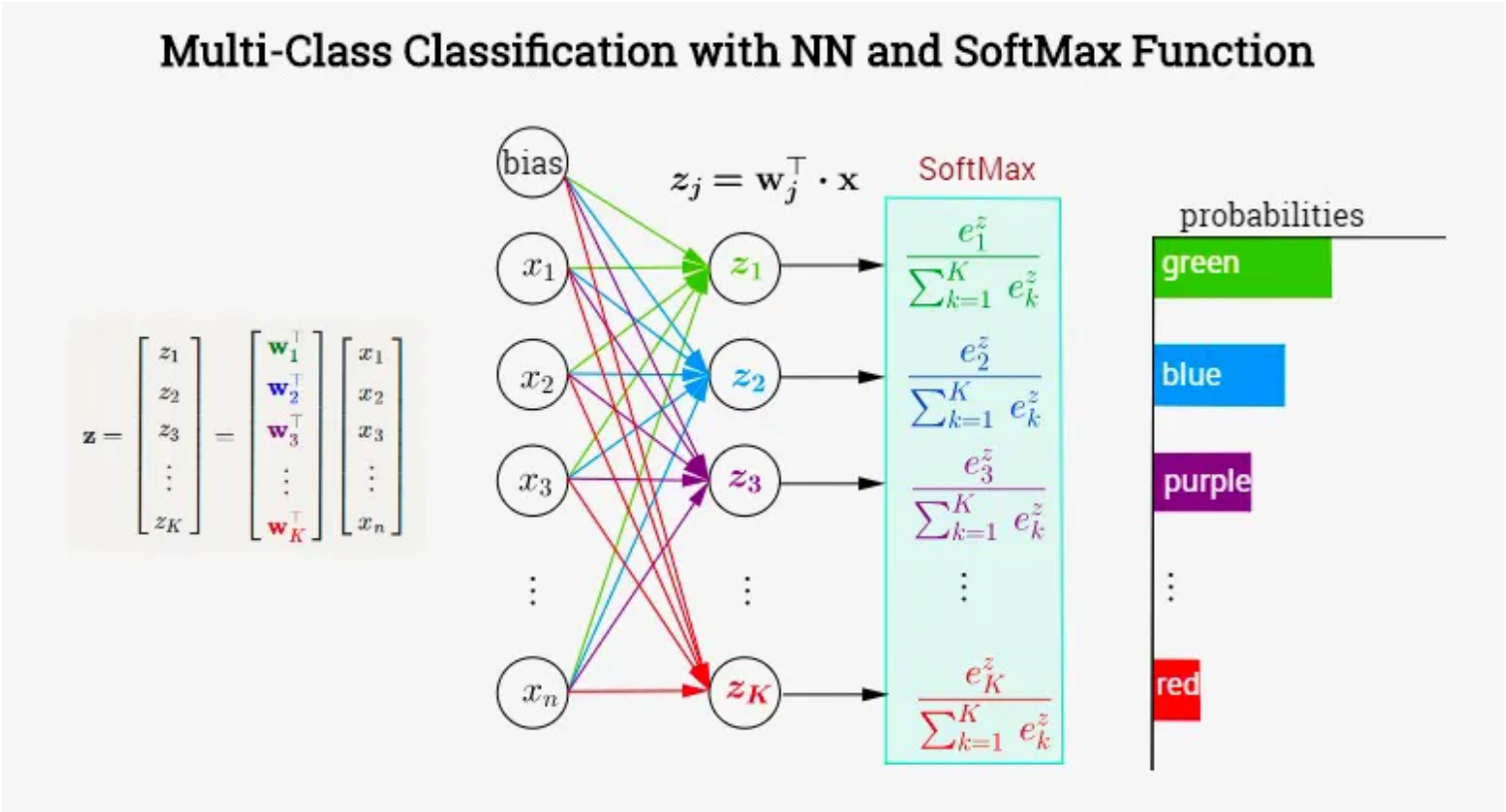
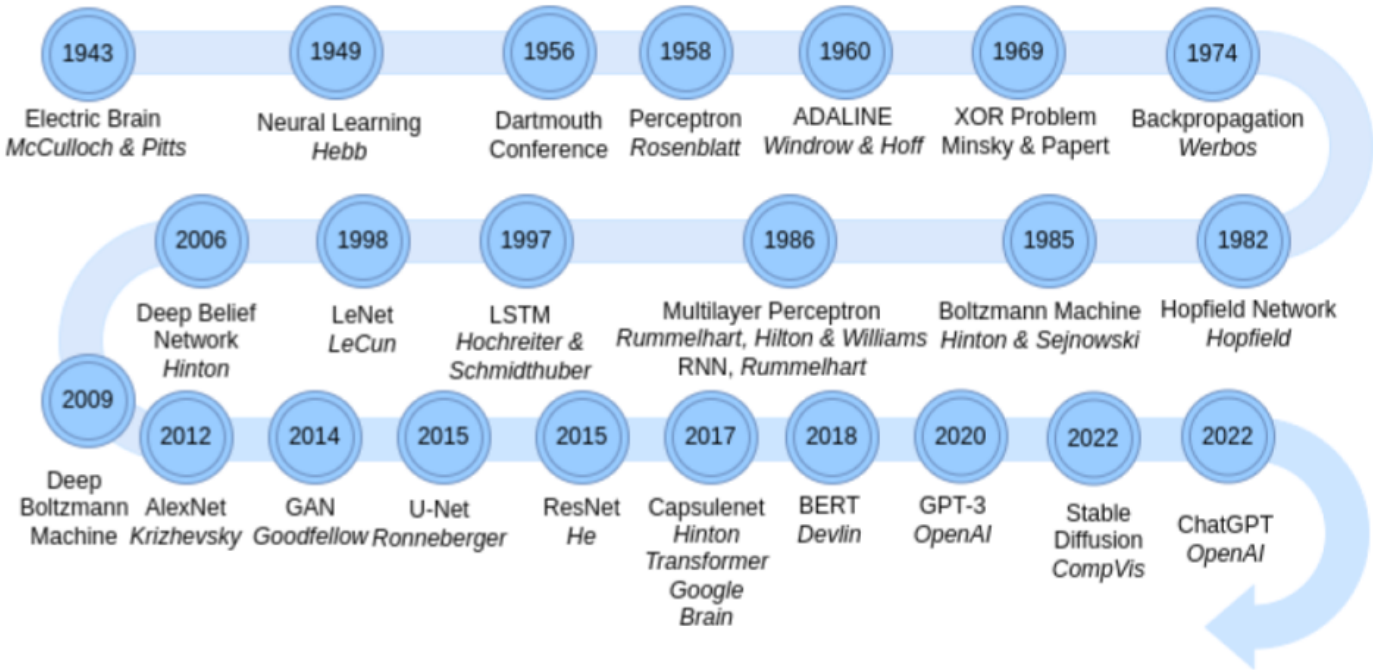


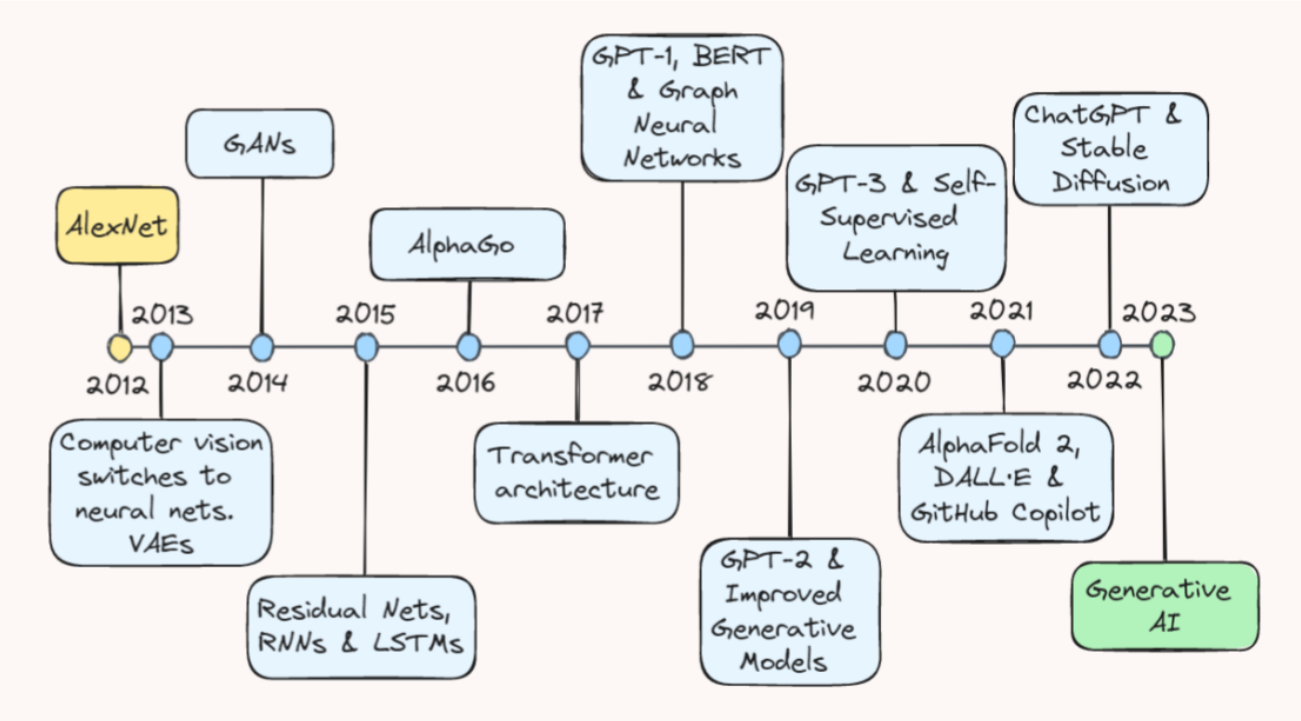
Figure: Softmax function Ref: `https://adeveloperdiary.com/data-science/deep-learning/neural-network-with-softmax-in-python/`

History of Neural Networks (1/2)



Ref: https://pub.towardsai.net/a-brief-history-of-neural-nets-472107bc2c9c

History of Neural Networks (2/2)



Neural network zoo:
`https://www.asimovinstitute.org/neural-network-zoo/`

Reference

- Abiodun, O.I., et. al.(2018).State-of-the-art in artificial neural network applications: A survey Heliyon, 4(11).